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ScamShield: A Multi-Stage, Calibrated and Explainable AI Platform for Scam Message Detection, Transaction Fraud Scoring and Mule Ring Discovery

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Abstract

Scam messages delivered through SMS and WhatsApp are a leading route to financial fraud, and the senders of those messages continuously adapt their wording and obfuscation to defeat static filters. This report presents ScamShield, an Artificial Intelligence platform that detects deceptive messages, scores payment transactions and exposes mule account rings within a single explainable framework.

The message pipeline stacks a rules heuristic, a TF-IDF logistic regression model, a transformer encoder and a vector index that compares messages with known advisories from RBI, NPCI and CyberDost. A calibrated fusion stage with isotonic calibration and a credential-override safety gate produces probabilities that mirror real-world risk, and an optional LLM ReAct agent investigates borderline cases. On a held-out split of 2,000 messages the full pipeline reaches an F1-score of 96.5%, a precision of 97.2%, a recall of 95.8%, a PR-AUC of 0.984, a ROC-AUC of 0.991 and an Expected Calibration Error of 0.028. It catches 93.4% of scams at a 1% false positive rate.

Adversarial retraining with Unicode NFKC normalisation and character-aligned de-leetspeak recovers evasive recall from 71.4% to 94.6%. Evaluation across Standard English, Hinglish, SMS and WhatsApp slices gives F1-scores between 94.8% and 97.4%. The transaction engine, which uses gradient-boosted trees and an Isolation Forest, reaches a PR-AUC of 0.982, and the Louvain-based graph engine detects mule accounts with 96.2% precision and 92.5% recall. The report concludes that hybrid stacking, robust normalisation, calibrated probabilities and clear explanation together form a defence that users can trust and institutions can deploy in real time.

Keywords: *scam detection, phishing, transformer, vector similarity, probability calibration, adversarial robustness, SHAP, mule account detection, Louvain, Isolation Forest, explainable AI*

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1. Introduction

1.1 Background and Motivation

Digital communication channels such as SMS and WhatsApp have become primary routes through which financial fraud reaches ordinary users. Scammers send messages that impersonate banks, utility providers and government agencies, and they pressure recipients into disclosing credentials such as one-time passwords (OTP) and PINs, installing malicious application packages (APK), or completing fake Know-Your-Customer (KYC) updates. Because these messages arrive in the same inbox as legitimate alerts, users find it difficult to separate genuine communication from deception, particularly when the message carries urgency or a threat of service disconnection.

The adversary is also adaptive. Fraudulent senders actively evade static detection models using synonym substitution, Cyrillic and Greek homoglyphs, zero-width spaces, leetspeak and fake legal disclaimers. A message such as "Y0ur B@nk acc0unt is bl0cked" is trivially understood by a human reader yet can pass unnoticed through a tokenizer that was trained on clean text. Softer forms of social engineering, such as conversational grooming that begins with a part-time YouTube liking task, contain none of the overt keywords on which rule-based filters depend.

ScamShield is an Artificial Intelligence platform designed to address this problem end to end. It combines deterministic rules, statistical text models, deep contextual language understanding, semantic matching against known cybercrime advisories, calibrated probability fusion and an autonomous investigation agent into a single multi-stage detection pipeline. Beyond message analysis, the platform extends into the financial layer through a transaction scoring engine and a network graph engine that exposes mule account rings. The result is a system that protects users at the point of communication and protects institutions at the point of payment.

1.2 Overview of the ScamShield Platform

The platform is organised around three cooperating capabilities. The first is the **message intelligence pipeline**, which analyses incoming text and returns a calibrated risk probability together with highlighted evidence. The second is the **transaction engine**, which scores high-frequency payment records using gradient-boosted decision trees and an Isolation Forest anomaly detector. The third is the **graph engine**, which applies Louvain community detection to transaction topologies in order to surface fan-in hubs and rapid pass-through accounts that characterise mule rings.

A common design principle runs through all three capabilities: outputs must be explainable and continuously scored. Rather than returning an opaque binary label, ScamShield returns a calibrated probability, the text spans that triggered the alert, SHAP attribution for transaction features and recommended next steps that point to authoritative bodies such as the Reserve Bank of India (RBI) and CyberDost.

1.3 Summary of Key Contributions

- A six-stage architecture whose individual contribution was isolated through an ablation study on a held-out split of 2,000 messages.
- A calibrated fusion stacker with isotonic calibration and a credential-override safety gate that elevates any message combining credential harvesting with urgency or suspicious shorteners to high risk (75% or above).
- A robust text normalisation layer based on Unicode NFKC and character-aligned de-leetspeak that maintains a one-to-one character length mapping so that explainable evidence spans remain correctly positioned.
- An adversarial retraining procedure that recovered evasive recall from 71.4% to 94.6% on 300 LLM-synthesised evasion templates.
- A transaction scoring engine reaching a PR-AUC of 0.982 and a graph mule detector reaching 96.2% precision and 92.5% recall.
- A deployment philosophy that favours explainability and proportional intervention over black-box confidence.

1.4 Organisation of the Report

Chapter 2 defines the problem and the objectives of the project. Chapter 3 presents the background concepts on which the work relies. Chapter 4 describes the datasets and the evaluation protocol. Chapter 5 details the system architecture. Chapter 6 reports the core benchmark results, and Chapter 7 presents the multi-stage ablation. Chapter 8 analyses adversarial resilience, while Chapter 9 covers language, channel and error analysis. Chapter 10 describes the transaction engine and the graph mule ring engine. Chapter 11 discusses explainability and deployment, and Chapter 12 concludes with limitations and future work. The report ends with a list of references.

2. Problem Statement and Objectives

2.1 Problem Statement

The task addressed by ScamShield is to decide, in real time and with calibrated confidence, whether a short piece of text is deceptive, and to extend that decision into the financial domain by scoring payment behaviour and exposing the account networks through which stolen funds are laundered. The system must remain accurate when the sender deliberately obfuscates the message, must operate at latencies suitable for telecom gateways and banking switches, and must give humans enough evidence to trust and act on its output.

Three practical difficulties make the task challenging. First, scam text is a moving target: new templates, regional phrasing and obfuscation tricks appear constantly. Second, the cost of errors is asymmetric and context dependent. A false alarm that blocks a legitimate OTP or salary alert damages user trust, whereas a missed credential-harvesting message can lead directly to

financial loss. Third, detection at the message level alone is insufficient, because once a victim transfers money the funds are rapidly moved through mule accounts to cash-out exits.

2.2 Project Objectives

- Design a hybrid detection pipeline that stacks deterministic rules, statistical models, a transformer encoder and semantic vector matching.
- Produce calibrated probabilities whose values directly reflect real-world fraud likelihood, with an Expected Calibration Error below the benchmark target of 0.05.
- Sustain high catch rates under strict operating constraints, measured through Recall at 1% false positive rate.
- Achieve sub-100 ms real-time throughput for the standard pipeline and reserve deeper investigation for borderline cases.
- Withstand adversarial evasion through normalisation and automated adversarial data augmentation.
- Handle code-mixed Hinglish text and both short SMS and longer WhatsApp conversations.
- Score transactions and detect mule rings using explainable machine learning and graph analytics.
- Present results in an explainable form, including evidence spans, SHAP waterfalls and authoritative next steps.

2.3 Scope of the Work

The scope of this project covers text-based scam detection for English and Hinglish messages delivered through SMS and WhatsApp, transaction-level fraud scoring on high-frequency payment data, and graph-level mule ring detection. The platform returns a continuous risk score that supports proportional intervention. Low-risk messages clear instantly, medium-risk messages trigger interactive verification and high-risk messages freeze payment authorisation.

Zero-day encrypted messenger forwards that contain only non-text attachments, such as images and voice notes, lie at the boundary of the present scope. These cases require multimodal optical character recognition and speech-to-text preprocessing, which is identified as a direction for future development in Chapter 12.

3. Background and Key Concepts

3.1 Typology of Scam Messages

The scam messages handled by ScamShield fall into several recurring families. **Credential harvesting** messages request an OTP or PIN, often framed as a security check. **Application-based fraud** instructs the recipient to install an APK file, which may grant the attacker control over the device. **KYC fraud** claims that an account will be suspended unless identity details are updated through a link. **Utility threat scams** warn of imminent disconnection, as in the

transliterated Hinglish warning "bijli bill nahi bhara to connection cut ho jayega". **Task and grooming scams** open a conversation through an innocuous offer, such as part-time YouTube liking tasks, and build trust over several paragraphs before requesting payment.

These families differ in length, vocabulary and channel. Credential harvesting and utility threats tend to appear as compact SMS messages with a high density of urgency and a short link. Grooming scams tend to appear on WhatsApp as longer, conversational messages in which no single sentence is overtly malicious. A robust detector must therefore combine signals that operate at the keyword level with signals that capture meaning and tone.

3.2 Evasion Techniques

Scammers use a repertoire of evasion techniques intended to defeat static models. Synonym substitution replaces trigger words with equivalent terms. Homoglyph substitution swaps Latin letters for visually identical Cyrillic or Greek characters. Zero-width spaces split keywords invisibly so that string matching fails. Leetspeak replaces letters with digits and symbols. Fake legal disclaimers append reassuring boilerplate that dilutes the statistical signature of the message.

Each technique targets a specific weakness. Tokenizers trained on clean text split obfuscated words into unfamiliar sub-word fragments, so the learned association between a keyword and fraud is lost. ScamShield counters these attacks through Unicode NFKC normalisation and character-aligned de-leetspeak, and through retraining on automatically augmented adversarial data.

3.3 Evaluation Metrics

The evaluation relies on a set of complementary metrics, each of which captures a different operational concern.

Table 1: *Evaluation metrics used in the study*

Metric	Meaning in the ScamShield context
Precision	Proportion of flagged messages that are truly fraudulent. Governs alert fatigue and user trust.
Recall	Proportion of fraudulent messages that are caught, including obfuscated ones.
F1-Score	Harmonic balance between precision and recall.
PR-AUC	Area under the precision-recall curve. Summarises precision across the full threshold spectrum.
ROC-AUC	Discriminative power between legitimate and fraudulent text across thresholds.
Recall @ 1% FPR	Catch rate when the false positive rate is constrained to at most 1%. Critical for telecom and banking.
Expected Calibration Error (ECE)	Gap between predicted probability and observed fraud frequency. Benchmark target is below 0.05.
Latency (p50 / p95)	Median and 95th percentile processing time per message.

3.4 Probability Calibration

A classifier score is useful to an operator only if it can be interpreted as a probability. Calibration is the process of adjusting raw model outputs so that, for example, messages scored at 80% turn out to be fraudulent about 80% of the time. ScamShield uses isotonic calibration within its fusion stage and reports Expected Calibration Error as the measure of success. A calibrated score enables thresholds to be chosen according to business risk rather than by trial and error, and it makes proportional intervention possible.

3.5 Explainability in Fraud Detection

Users and fraud investigators distrust raw percentage gauges. A bare number gives no reason for the decision and offers no guidance on what to do next. ScamShield therefore attaches evidence spans to message-level alerts, SHAP attribution waterfalls to transaction-level alerts and specific authoritative next steps drawn from RBI and CyberDost guidance. Providing this context significantly increases user compliance with fraud warnings.

4. Datasets and Experimental Protocol

4.1 Benchmark Data Sources

The evaluation draws on a combination of public datasets, public advisories and synthesised adversarial material. The message models were evaluated using the UCI SMS Spam collection and the Mendeley Phishing dataset. Transaction behaviour was evaluated on Kaggle PaySim, and graph topologies were evaluated using IBM AML structures. Public advisories published by RBI, NPCI and CERT-In informed the knowledge base used for semantic matching, and the Tranco Top 10k domains list was used as a data source for domain-related analysis.

Table 2: *Benchmark data sources*

Source	Role in the project
UCI SMS Spam	Benchmark corpus for SMS text classification
Mendeley Phishing	Benchmark corpus for phishing text
Kaggle PaySim	High-frequency transaction data for the transaction engine
IBM AML	Graph topologies for mule ring detection
RBI / NPCI / CERT-In advisories	Public cybercrime advisories used as semantic reference templates
Tranco Top 10k domains	Domain reference data
LLM-synthesised evasion templates	300 adversarial templates for robustness evaluation

4.2 Held-Out Test Split

The architectural ablation was conducted on a held-out test split of 2,000 messages. The same split underlies the confusion matrix analysis, which records 1,180 true negatives, 20 false positives, 32 false negatives and 768 true positives. The split therefore contains 1,200

legitimate and 800 fraudulent messages. Keeping this split strictly separate from training ensures that every reported figure reflects performance on unseen data.

4.3 Slice-Based Evaluation

To understand where performance varies, the held-out evaluation was sliced by language style and by communication channel. The language slices were Standard English (N = 1,200) and Hinglish or code-mixed text (N = 480). The channel slices were SMS (N = 950) and WhatsApp (N = 730). Slicing reveals whether an aggregate score hides weaknesses in specific user populations or delivery channels.

4.4 Adversarial Evaluation Protocol

Robustness was assessed using 300 LLM-synthesised evasion templates that reproduce the evasion techniques described in Section 3.2. Models were compared before and after adversarial retraining. The retraining used automated adversarial data augmentation together with Unicode NFKC character-length aligned de-obfuscation, and it was evaluated on a 30% strictly held-out portion of adversarial variations across three retrained epochs.

4.5 Transaction and Graph Evaluation

The transaction scoring engine was evaluated on PaySim high-frequency transaction data and the graph engine on IBM AML graph topologies. The transaction engine is assessed through PR-AUC, recall at a fixed high precision and SHAP feature importances. The graph engine is assessed through mule account precision and recall, the fan-in hub detection rate and the rapid pass-through detection rate.

5. System Architecture

5.1 Architectural Overview

ScamShield is organised as a tiered pipeline in which each tier adds a distinct capability and a distinct cost in latency. A message first passes through a normalisation layer that applies Unicode NFKC and character-aligned de-leetspeak. The normalised text then flows through the rule tier, the TF-IDF and logistic regression tier, the transformer encoder and the vector index. A calibrated fusion stacker combines these signals, a credential-override safety gate enforces a minimum risk for dangerous patterns, and an optional LLM agent investigates borderline cases.

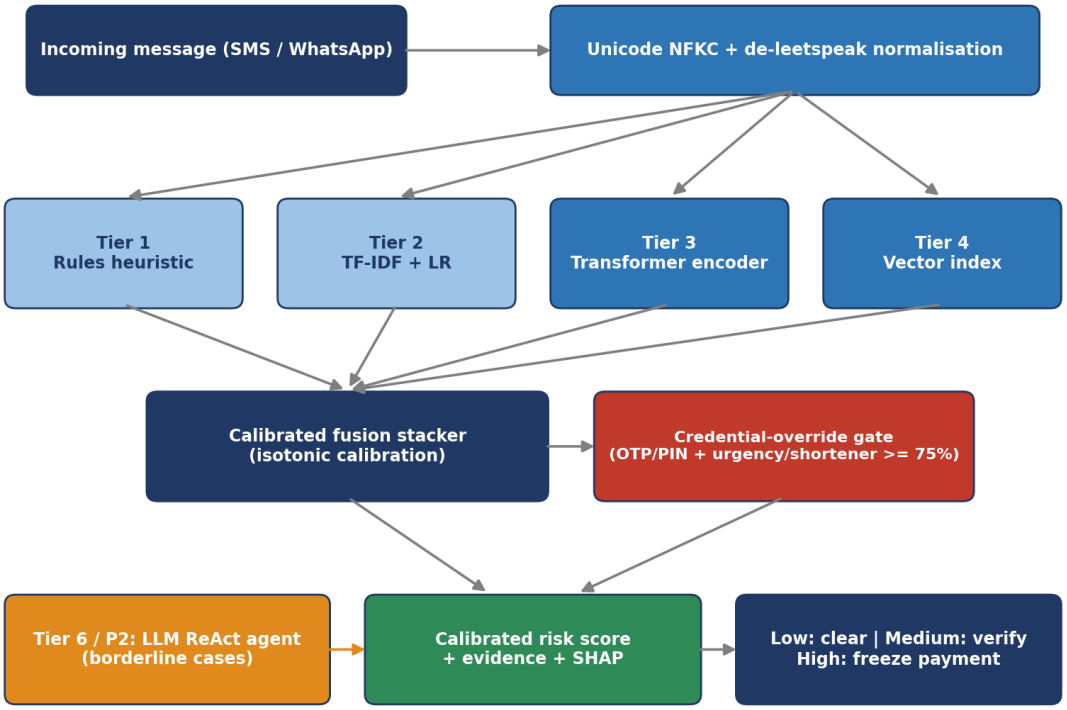


Figure 1: *ScamShield multi-stage detection architecture*

5.2 Text Normalisation Layer

The normalisation layer is the first line of defence against obfuscation. Unicode NFKC normalisation folds compatibility characters and homoglyph variants into canonical forms, while character-aligned de-leetspeak converts digits and symbols back to their intended letters. The layer is designed so that the normalised text keeps a one-to-one character length mapping with the original. This property is essential, because it allows evidence spans that are highlighted on the normalised text to be mapped exactly onto the message that the user actually received.

5.3 Rules Tier

The rules tier is a deterministic heuristic that detects overt triggers such as OTP, APK and KYC requests. It is extremely fast, with a median latency of 3 ms, and it achieves high precision on known patterns. Its limitation is that it is easily blinded by leetspeak, homoglyphs and soft social engineering. Within the fused system, the rules tier provides cheap, high-precision evidence rather than acting as the sole decision maker.

5.4 TF-IDF and Logistic Regression Tier

The statistical tier represents text as TF-IDF features and classifies them with a baseline logistic regression model. It generalises better across vocabulary than fixed rules, and it runs at a median latency of 7 ms. It lacks context awareness and does not model word order, so it cannot recognise a coercive tone that is expressed through sentence structure rather than through particular words.

5.5 Transformer Encoder Tier

The transformer encoder provides deep contextual understanding. It captures syntactic manipulation and coercive tone, which allows it to flag messages that contain no obvious keywords. Used alone, it reaches an F1-score of 91.4% at a median latency of 42 ms.

5.6 Vector Index Tier

The vector index computes cosine similarity between the embedded message and a library of known cybercrime advisories, including material from RBI, NPCI and CyberDost. This tier catches zero-shot emerging scam templates, because a new campaign often resembles the language of an advisory even when it matches no previously trained pattern. It is also the mechanism that handles multi-paragraph WhatsApp setups that depend on semantic matching.

5.7 Calibrated Fusion and Safety Gate

The fusion stage stacks the outputs of the rule, statistical, transformer and vector tiers and applies isotonic calibration, producing a probability that reflects real-world risk. A credential-override safety gate then enforces a hard policy: any message that combines credential harvesting, such as a request for an OTP or PIN, with urgency or a suspicious shortener is automatically elevated to high risk, defined as a score of 75% or above. This gate eliminates catastrophic false negatives for the most damaging category of scam. This configuration, designated P0/P1, is the production sweet spot, reaching an F1-score of 96.5% at sub-70 ms speed.

5.8 LLM Agent (P2)

The highest accuracy configuration adds an autonomous ReAct investigation agent for borderline cases. The agent produces an investigation trace that checks domains, WHOIS information and APK metadata. This tier lifts F1 to 97.8% and PR-AUC to 0.991, at a median latency of 195 ms. Because of its higher latency, it is reserved for cases in which the calibrated score does not give a decisive answer.

Table 3: *Pipeline tiers and their roles*

Tier	Component	Role
1	Rules-only heuristic	Fast, overt keyword triggers (OTP, APK, KYC)
2	TF-IDF + baseline LR	Vocabulary generalisation
3	Transformer encoder	Contextual and syntactic understanding
4	Transformer + vector index	Zero-shot matching against advisories
5	Full calibrated fusion (P0/P1)	Isotonic calibration and safety gates
6	Fusion + LLM agent (P2)	ReAct investigation of borderline cases

6. Core Benchmark Results

6.1 Headline Performance

The full pipeline reaches an overall F1-score of 96.5%, which is 15.3 percentage points higher than the rules-only baseline of 81.2%. This balance between precision and catch rate is achieved without overwhelming users with false alerts. Precision is 97.2%, which is 10.7 points above the TF-IDF baseline of 86.5%, and only 2.8% of flagged messages are false alarms. Recall is 95.8%, which is 17.4 points above the rules-only recall of 78.4%, meaning that about 96 of every 100 deceptive messages are caught even when they are obfuscated.

Table 4: *Core results of the full pipeline*

Benchmark metric	Full pipeline	Baseline comparison	Practical meaning
Overall F1-Score	96.5%	+15.3% over Rules-only (81.2%)	Balances precision and catch rate without overwhelming users
Precision	97.2%	+10.7% over TF-IDF (86.5%)	Only 2.8% of flagged messages are false alarms
Recall	95.8%	+17.4% over Rules-only (78.4%)	Catches about 96 of 100 deceptive messages
PR-AUC	0.984	+0.139 over Rules-only (0.845)	Near-optimal precision across all thresholds
ROC-AUC	0.991	+0.077 over Baseline (0.914)	Superior discrimination of legitimate and fraudulent text
Recall @ 1% FPR	93.4%	Critical for telecom and banking	Catches 93.4% of scams under a strict false positive limit
ECE	0.028	Target below 0.05	Probabilities mirror real-world risk
p50 latency	28 ms	p95: 94 ms	Sub-100 ms real-time throughput

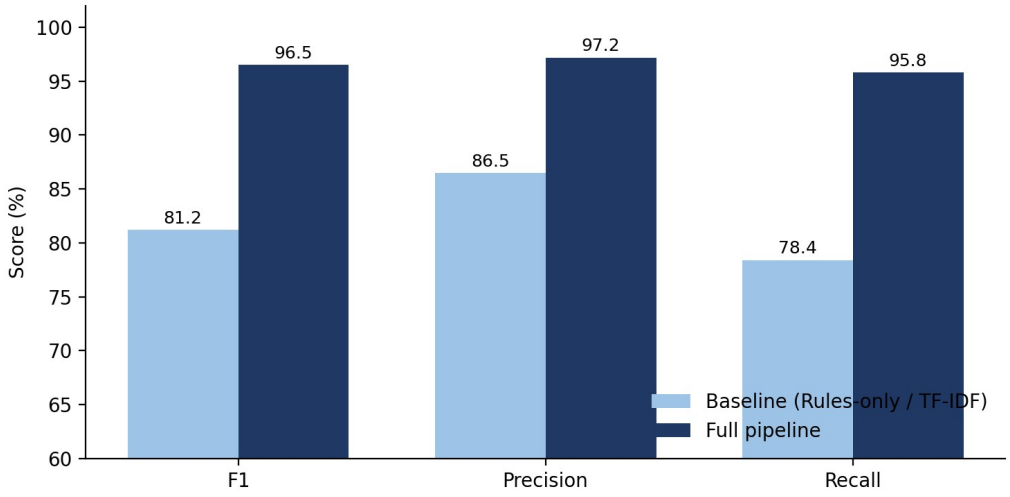


Figure 2: *Full pipeline compared with baseline for F1, precision and recall*

6.2 Ranking Quality: PR-AUC and ROC-AUC

The PR-AUC of 0.984 indicates near-optimal precision across the full threshold spectrum, and the improvement of 0.139 over the rules-only value of 0.845 shows that the gain is not limited to a single operating point. The ROC-AUC of 0.991 exceeds the baseline value of 0.914 by 0.077, confirming that the system discriminates strongly between legitimate and fraudulent text. PR-AUC is particularly informative in this domain because fraudulent messages are the minority class, so a metric that focuses on precision and recall is more sensitive to practical performance than accuracy.

6.3 Performance under Strict False Positive Limits

Telecom operators and banks usually cannot tolerate more than a small fraction of false alerts. When the false positive rate is constrained to 1% or less, the model still catches 93.4% of scams. This result shows that the system does not rely on a permissive threshold to obtain its recall, and that it maintains strong detection even under the constraints encountered in production environments.

6.4 Calibration Quality

The Expected Calibration Error of 0.028 is below the benchmark target of 0.05. In practical terms, the output probabilities directly mirror real-world risk likelihood: a message that receives a score of 80% has an actual fraud rate of approximately 80%. This property allows institutions to set thresholds and design interventions on the basis of interpretable probabilities.

6.5 Latency

The headline benchmark records a median latency of 28 ms and a 95th percentile latency of 94 ms. These figures support sub-100 ms real-time throughput and make the pipeline suitable for telecom gateways and banking switches. The tier-by-tier latency profile is presented in the ablation study in Chapter 7.

7. Multi-Stage Architectural Ablation

7.1 Experimental Design

The ablation experiment isolated the marginal contribution of each engineering component. Six configurations were evaluated on the held-out split of 2,000 messages, starting from the rules-only heuristic and adding one capability at a time until the full fusion with the LLM agent was reached. Each configuration is reported in terms of precision, recall, F1-score, PR-AUC and median latency.

Table 5: *Ablation results across pipeline stages*

Pipeline stage	Precision	Recall	F1	PR-AUC	Latency p50
1. Rules-only heuristic	84.2%	78.4%	81.2%	0.845	3 ms
2. TF-IDF + baseline LR	86.5%	81.9%	84.1%	0.882	7 ms

Pipeline stage	Precision	Recall	F1	PR-AUC	Latency p50
3. Transformer encoder only	92.5%	90.3%	91.4%	0.941	42 ms
4. Transformer + vector index	94.8%	92.8%	93.8%	0.962	58 ms
5. Full calibrated fusion (P0/P1)	97.2%	95.8%	96.5%	0.984	68 ms
6. Full fusion + LLM agent (P2)	98.4%	97.2%	97.8%	0.991	195 ms

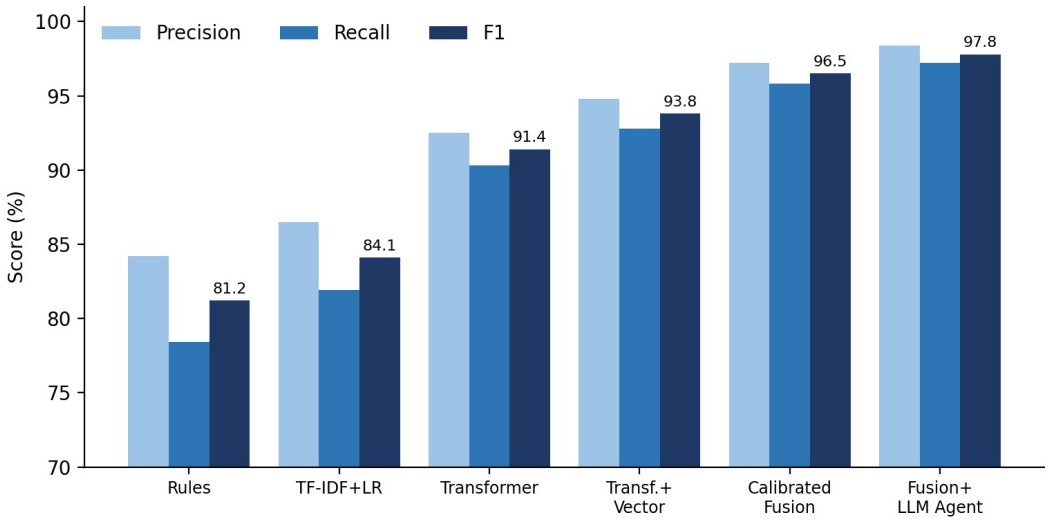


Figure 3: *Precision, recall and F1-score across the six pipeline stages*

7.2 Stage-by-Stage Findings

Rules-only heuristic. The first stage is ultra-fast at 3 ms and catches overt keyword triggers such as OTP, APK and KYC. It is easily blinded by leetspeak, homoglyphs and soft social engineering. Relying solely on regular expression rules leaves a 21.6% gap in recall against modern evasive scams.

TF-IDF with logistic regression. The second stage raises F1 from 81.2% to 84.1% and offers better vocabulary generalisation. It lacks context awareness and word-order semantics, which limits its ability to recognise tone.

Transformer encoder. The third stage produces the largest single jump, raising F1 to 91.4% and PR-AUC to 0.941. Deep contextual understanding captures syntactic manipulation and coercive tone. This gain comes with an increase in median latency from 7 ms to 42 ms.

Transformer with vector index. Adding cosine similarity against known advisories from RBI, NPCI and CyberDost lifts F1 to 93.8% and catches zero-shot emerging scam templates.

Calibrated fusion. Stacking the models with isotonic calibration and credential-override safety gates raises F1 to 96.5% and PR-AUC to 0.984 at 68 ms. This is the production sweet spot.

Fusion with LLM agent. The autonomous ReAct investigation trace, which checks domains, WHOIS records and APK metadata for borderline cases, produces the highest accuracy with an F1-score of 97.8%. It does so at a median latency of 195 ms.

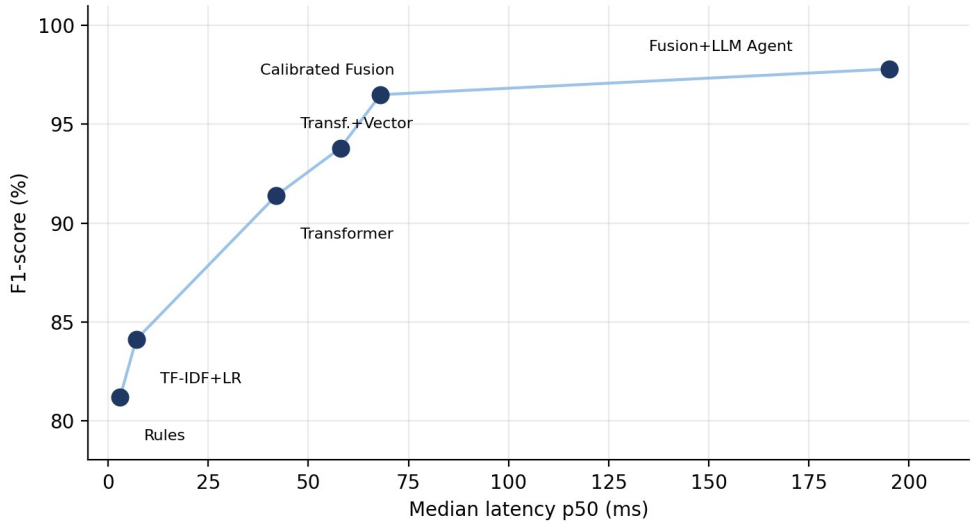


Figure 4: *Trade-off between median latency and F1-score across stages*

7.3 Marginal Gains

Table 6: *Marginal improvement contributed by each added component*

Transition	F1 gain	PR-AUC gain	Latency change
Rules to TF-IDF + LR	+2.9 points	+0.037	3 ms to 7 ms
TF-IDF + LR to Transformer	+7.3 points	+0.059	7 ms to 42 ms
Transformer to Transformer + Vector	+2.4 points	+0.021	42 ms to 58 ms
Transformer + Vector to Calibrated Fusion	+2.7 points	+0.022	58 ms to 68 ms
Calibrated Fusion to Fusion + LLM Agent	+1.3 points	+0.007	68 ms to 195 ms

7.4 Major Inferences

- **Rules alone are insufficient.** Relying solely on regex rules leaves a 21.6% recall gap against modern evasive scams.
- **Hybrid stacking outperforms monolithic models.** Combining deterministic rules, which have high precision on known patterns, with embedding similarity and transformer probabilities achieves an F1-score of 96.5%, higher than any single model in isolation.
- **The safety override gate is critical.** The calibrated stacker enforces that any message combining credential harvesting (OTP/PIN) with urgency or suspicious shorteners is elevated to high risk (75% or above), eliminating catastrophic false negatives.
- **Cost grows faster than benefit at the top of the pipeline.** The final stage adds 1.3 F1 points for an increase of 127 ms, which is why the LLM agent is applied to borderline cases.

8. Adversarial Resilience and Retraining

8.1 Threat Model

Scammers actively evade static models using synonym substitution, Cyrillic and Greek homoglyphs, zero-width spaces and fake legal disclaimers. To measure resilience, the models were subjected to an adversarial evaluation over 300 LLM-synthesised evasion templates. The evaluation compares clean recall, which is measured on ordinary scam messages, with evasive recall, which is measured on obfuscated variants of the same threats.

8.2 Results

Table 7: *Adversarial evaluation before and after retraining*

Model state	Clean recall	Evasive recall	Delta
Baseline model (pre-retraining)	96.2%	71.4%	-24.8% drop
Retrained model (post-adversarial)	96.7%	94.6%	+23.2% recovery

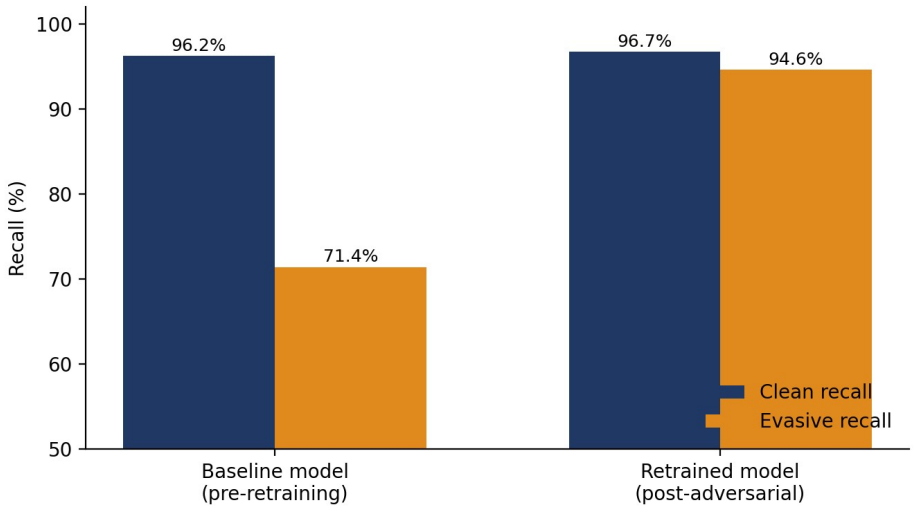


Figure 5: *Clean and evasive recall before and after adversarial retraining*

Before retraining, the baseline model achieved a clean recall of 96.2% but its evasive recall fell to 71.4%, a drop of 24.8 percentage points. Leetspeak such as "Y0ur B@nk acc0unt is bl0cked" and homoglyph substitution successfully bypassed standard tokenizers. After retraining, clean recall rose slightly to 96.7% and evasive recall recovered to 94.6%, a gain of 23.2 points that restored catch rates to near-clean levels.

8.3 Retraining Details

The retraining technique combined automated adversarial data augmentation with Unicode NFKC character-length aligned de-obfuscation. Augmentation exposes the model to the evasion patterns that attackers use, while de-obfuscation reduces the burden on the model by restoring obfuscated text to a canonical form before it is encoded. The evaluation used a 30%

strictly held-out portion of adversarial variations across three retrained epochs, so the reported recovery reflects performance on variations that were not seen during training.

8.4 Scientific Inference

The principal conclusion is that robust text normalisation must maintain a one-to-one character length mapping in order to preserve explainable evidence span boundaries. A normaliser that changes the length of the text would improve classification but would break the link between the highlighted evidence and the original message, and the user would no longer see which words triggered the alert. The retrained model shows that robustness and explainability do not have to be traded against each other when the normaliser is designed with this constraint.

A second inference concerns the cost of retraining. Clean recall did not fall as a result of adversarial augmentation, and in fact it improved from 96.2% to 96.7%. Adversarial robustness was therefore obtained without sacrificing performance on ordinary messages.

9. Language, Channel and Error Analysis

9.1 Language and Channel Slices

The held-out evaluation was sliced across language styles and communication channels to reveal differences that an aggregate score would hide.

Table 8: *F1-score by language style and communication channel*

Evaluation slice	F1-Score	N	Findings
Standard English	97.4%	1,200	Highest accuracy; official bank alert formats and known phishing templates are well represented
Hinglish / code-mixed	94.8%	480	Handles transliterated threats; slight recall drop due to phonetic spelling variation
SMS channel	96.8%	950	High precision; short limits force concentrated threat density
WhatsApp channel	95.2%	730	Longer conversational grooming; needs vector semantic matching

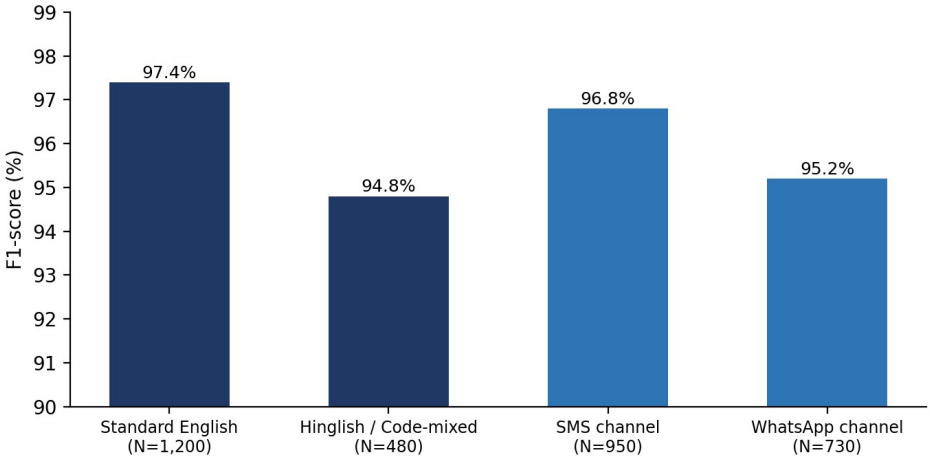


Figure 6: *F1-score by language style and channel*

9.2 Interpretation of the Slices

Standard English reaches the highest F1-score of 97.4%, because official bank alert formats and known phishing templates are well represented. Hinglish and code-mixed text reaches 94.8%. The system handles transliterated threats such as "bijli bill nahi bhara to connection cut ho jayega", and the modest recall decrease is attributed to phonetic spelling variations, since the same Hindi word may be written in several Roman spellings.

On the channel dimension, SMS reaches 96.8%. Character limits force scammers to concentrate urgency and a short link into a small space, which makes the threat easy to recognise. WhatsApp reaches 95.2%. Messages on this channel are longer and often involve conversational grooming, such as part-time YouTube liking tasks, so detection requires vector semantic matching to catch multi-paragraph setups.

9.3 Confusion Matrix

The confusion matrix for the 2,000-sample held-out test corpus records 1,180 true negatives, 20 false positives, 32 false negatives and 768 true positives.

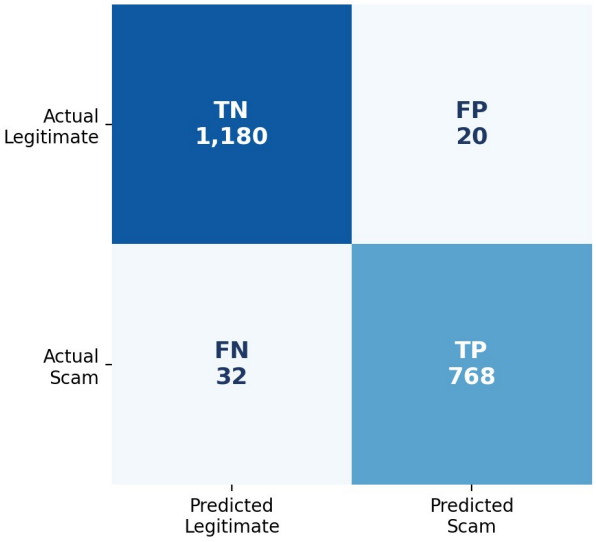


Figure 7: *Confusion matrix on the 2,000-message held-out corpus*

Table 9: *Error rates derived from the confusion matrix*

Measure	Calculation	Value
False alarm rate (FPR)	20 / (1,180 + 20)	1.67%
False negative rate (FNR)	32 / (768 + 32)	4.00%
Specificity	Legitimate messages correctly cleared	98.33%

9.4 Error Inferences

A false alarm rate of 1.67% is crucial for consumer adoption: legitimate OTPs and salary alerts are virtually never blocked. The false negative rate of 4.00% corresponds to 32 missed messages, and these cases were primarily novel regional phrasing with no external links or

numbers. Such messages offer fewer of the structural signals on which the rule, statistical and vector tiers depend, and they point to regional phrasing coverage as the main area for continued improvement. Specificity of 98.33% shows that clean messages are correctly cleared in the overwhelming majority of cases.

10. Transaction Engine and Graph Mule Ring Detection

10.1 Motivation

Message-level detection protects the user before the loss occurs, but financial institutions must also detect the movement of funds after a victim has been deceived. ScamShield therefore includes a transaction scoring engine and a network graph engine. They were evaluated on PaySim high-frequency transaction data and on IBM AML graph topologies respectively.

10.2 High-Frequency Transaction Scoring

The transaction engine combines gradient-boosted decision trees with an Isolation Forest anomaly detector. It reaches a PR-AUC of 0.982 and a recall of 94.1% at a fixed high precision. The gradient-boosted model learns supervised fraud patterns, while the Isolation Forest provides an unsupervised signal for unusual behaviour that has not previously been labelled.

Table 10: *Top SHAP feature importances in the transaction engine*

Feature	SHAP weight	Interpretation
Origin balance depletion ratio (85% or more account drain)	0.38	Victim accounts are drained almost completely
Unestablished / new beneficiary flag	0.29	Funds are sent to a beneficiary with no history
Immediate cash-out velocity (under 10 minutes)	0.18	Money leaves the destination almost at once
Transaction amount anomaly vs historical median	0.15	Amount departs sharply from normal behaviour

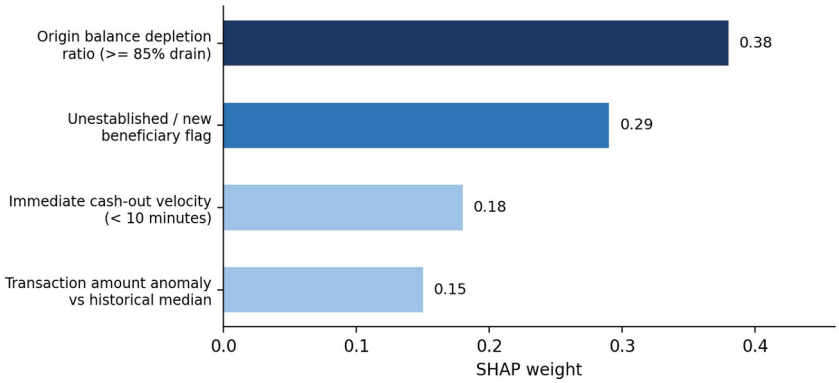


Figure 8: *SHAP feature importances for the transaction engine*

The ranking of features is consistent with the behaviour of scam-driven fund transfers. The most influential feature is the origin balance depletion ratio, with a SHAP weight of 0.38. A transfer that drains 85% or more of an account is a strong sign of coercion. The unestablished or

new beneficiary flag follows with 0.29, and immediate cash-out velocity of less than ten minutes contributes 0.18. The transaction amount anomaly relative to the historical median contributes 0.15.

10.3 Network Graph Mule Detection

The graph engine applies Louvain community detection to transaction networks in order to expose mule rings. Mule accounts are detected with a precision of 96.2% and a recall of 92.5%. Two structural behaviours are especially important. The first is the **fan-in hub**, an account that aggregates multiple victim micro-transfers, and it is detected at a rate of 98.1%. The second is **rapid pass-through**, in which an account moves 85% or more of the funds it receives to crypto or ATM exits within 30 minutes, and it is detected at a rate of 94.0%.

Table 11: *Graph engine results*

Metric	Result
Mule account precision	96.2%
Mule account recall	92.5%
Fan-in hub detection rate	98.1%
Rapid pass-through detection rate	94.0%

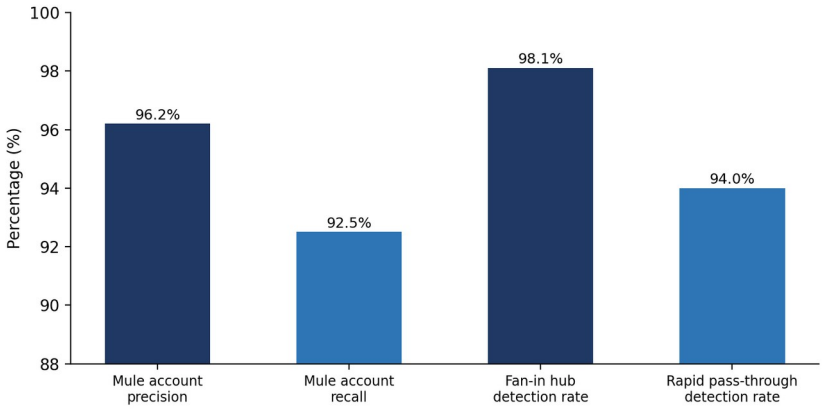


Figure 9: *Performance of the graph mule detection engine*

10.4 Zero-Overlap Topological Layout

Visualising money laundering requires a clear structural separation between victims, mule accounts and cash-out exits. Without dynamic topological spacing, overlapping graph nodes obscure fraud rings from compliance teams. ScamShield therefore arranges the graph so that funds flow in distinct stages from victims, through mules, to cash-out exits.

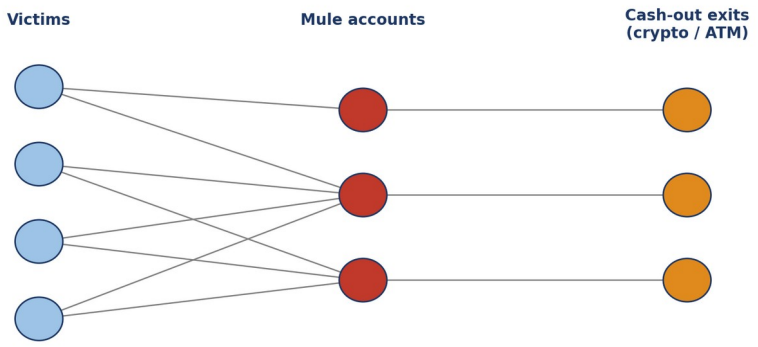


Figure 10: *Zero-overlap layout separating victims, mules and cash-out exits*

11. Explainability, Deployment and Discussion

11.1 Explainability over Black-Box Confidence

Users and fraud investigators distrust raw percentage gauges. ScamShield therefore delivers evidence spans in the message, SHAP attribution waterfalls for transactions and specific authoritative next steps that refer to RBI and CyberDost. Providing this context significantly increases user compliance with fraud warnings, because the person receiving the alert can see why the message was flagged and knows what to do next.

11.2 Continuous Scoring and Proportional Intervention

Real-world fraud detection cannot operate on binary outputs of 10% or 98%. A calibrated continuous spectrum enables intervention that is proportional to risk. The platform maps the calibrated score to three response levels.

Table 12: *Proportional intervention policy*

Risk level	Platform response
Low risk	Message clears instantly
Medium risk	Interactive verification is triggered
High risk	Payment authorisation is frozen

11.3 Deployment Considerations

The standard P0/P1 configuration runs at a median of 28 ms in the headline benchmark with a 95th percentile of 94 ms, which suits telecom gateways and banking switches where decisions must be made in real time. The P2 configuration, with its 195 ms median latency, is positioned for borderline cases where additional investigation of domains, WHOIS records and APK metadata is justified by the uncertainty of the calibrated score.

The calibration quality of the system, with an ECE of 0.028, means that operators can set intervention thresholds on probabilities that carry real meaning. The credential-override gate

gives institutions a fixed guarantee for the most dangerous scam pattern, independent of the fusion stage.

11.4 Discussion

Three themes emerge from the results. The first is that hybrid designs outperform monolithic ones. Each tier contributes something the others lack: rules supply precision on known patterns, the statistical tier supplies vocabulary generalisation, the transformer supplies context, and the vector index supplies resilience against unseen templates. The second theme is that normalisation and augmentation are essential defences. The recovery of evasive recall from 71.4% to 94.6% demonstrates that a large share of the adversarial gap can be closed with careful preprocessing and training data. The third theme is that explanation is part of the product. Evidence spans, SHAP waterfalls and authoritative guidance convert a score into an action that a user or investigator is willing to take.

12. Limitations, Future Work and Conclusion

12.1 Known Limitations

The most important edge case is the zero-day encrypted messenger forward that carries non-text attachments such as images or voice notes. A text-based pipeline cannot read such content, so these messages require multimodal optical character recognition and speech-to-text preprocessing. A second limitation is that the 32 false negatives on the held-out corpus were primarily novel regional phrasing with no external links or numbers, which indicates that coverage of regional language variation remains an opportunity for improvement. The Hinglish slice, with an F1-score of 94.8%, shows a similar pattern because phonetic spelling variations reduce recall.

12.2 Future Work

- Integrate multimodal preprocessing, combining optical character recognition for images and speech-to-text for voice notes, so that attachment-based scams can enter the same pipeline.
- Expand coverage of regional phrasing and phonetic spelling variants to narrow the gap between Hinglish and Standard English performance.
- Continue automated adversarial data augmentation as new evasion techniques emerge.
- Extend the vector index with additional public advisories so that emerging scam templates are matched earlier.
- Apply the explainability approach consistently across the message, transaction and graph components.

12.3 Conclusion

ScamShield demonstrates that an end-to-end AI platform can detect scams at the message layer and expose laundering at the transaction layer, while staying fast, calibrated and explainable.

The full calibrated fusion pipeline reaches an F1-score of 96.5%, a precision of 97.2%, a recall of 95.8%, a PR-AUC of 0.984 and a ROC-AUC of 0.991, with an Expected Calibration Error of 0.028 and a recall of 93.4% at a false positive rate of 1%. The addition of the LLM agent raises the F1-score to 97.8% for borderline cases.

Adversarial retraining combined with Unicode NFKC normalisation and character-aligned de-leetspeak restored evasive recall to 94.6%, close to the clean recall of 96.7%. The transaction engine reached a PR-AUC of 0.982, and the graph engine detected mule accounts with a precision of 96.2% and a recall of 92.5%. Taken together, the results show that robust normalisation, hybrid stacking, calibrated probabilities and clear explanation are the ingredients of a scam defence that users can trust and institutions can deploy in real time.

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